

Stomach, liver, and intestines



ANCIENT HUMORS

In ancient times a popular theory, adopted by physicians for centuries afterward, involved four fluids, or humors, contained by the body: blood, mucus (phlegm), black bile, and yellow bile. The balance of the humors determined a person's temperament. Blood, or *Sanguis*, made a person sanguine, or cheerful; phlegm, or *Pituita*, was phlegmatic, or slow-moving; yellow bile, or *Chole*, was choleric, or hot-tempered; black bile, or *Melanchole*, was melancholic, or depressed.

THE LIVER'S CHEMICAL "FACTORY"

The body's largest internal organ is the liver. Soft, and a rich dark-red with blood, it has a large right lobe and a much smaller left lobe. Blood flows to the liver directly from the heart, and also from the intestines. The blood oozes through some 50,000 tiny units called hepatic lobules, about 0.04 in (1 mm) wide. Here, chemical processes break down digested nutrients even further, assemble useful substances, filter old cells from blood, and store or release sugars, starches, fats, vitamins, and minerals according to the body's changing needs.

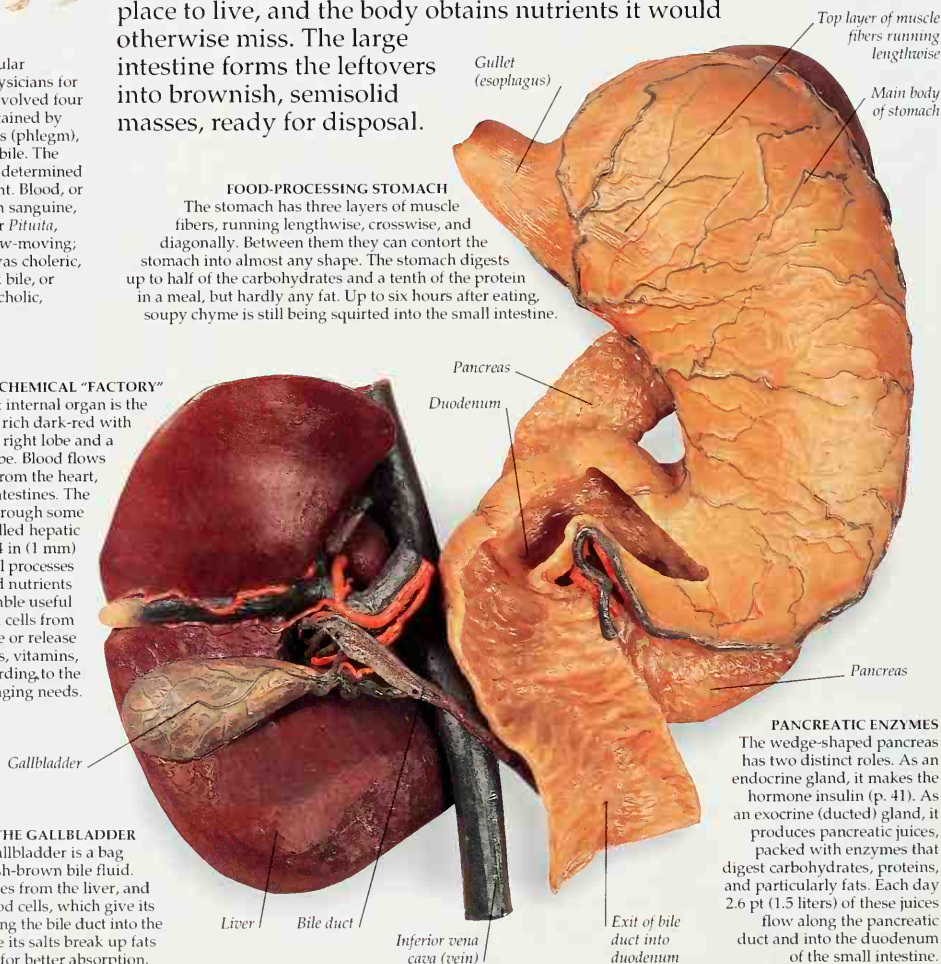
FATS, BILE, AND THE GALLBLADDER

The soft, elastic gallbladder is a bag that stores greenish-brown bile fluid. Bile contains wastes from the liver, and broken-down blood cells, which give its color. It passes along the bile duct into the duodenum, where its salts break up fats into tiny droplets for better absorption.

AFTER FOOD IS SWALLOWED IT TRAVELS in soft lumps down the gullet into the stomach. The stomach stretches as it fills, and can hold 3.5 pt (2 liters) or more. The stomach is both a food mixer and acid-and-enzyme bath. Every few minutes its strong muscular walls undergo a spasm of squeezing to churn the food into a semiliquid called chyme. Only a few substances are actually absorbed through the stomach lining into the blood. The next stage of the digestive tube is the small intestine, where chemical digestion continues, and most nutrients are absorbed. Inside a healthy intestine live friendly bacteria. These can make important nutrients such as vitamin K, can help break down thick cellulose walls of plant cells, and are involved in processing bile salts. Both members of this partnership benefit – the bacteria have a warm, food-rich place to live, and the body obtains nutrients it would otherwise miss. The large intestine forms the leftovers into brownish, semisolid masses, ready for disposal.

FOOD-PROCESSING STOMACH

The stomach has three layers of muscle fibers, running lengthwise, crosswise, and diagonally. Between them they can contort the stomach into almost any shape. The stomach digests up to half of the carbohydrates and a tenth of the protein in a meal, but hardly any fat. Up to six hours after eating, soupy chyme is still being squirted into the small intestine.



PANCREATIC ENZYMES

The wedge-shaped pancreas has two distinct roles. As an endocrine gland, it makes the hormone insulin (p. 41). As an exocrine (ducted) gland, it produces pancreatic juices, packed with enzymes that digest carbohydrates, proteins, and particularly fats. Each day 2.6 pt (1.5 liters) of these juices flow along the pancreatic duct and into the duodenum of the small intestine.